Contents

Operator HOW-TO — ESP32-S3 IMU, phone, and MT200

Audience: field operator (not firmware developer).

Kit: Waveshare **ESP32-S3-LCD-1.47B** (QMI8658 IMU), Android app **ESP32S3 IMU sim**, optional **Veepoo / H-Band MT200** watch, optional cloud **Good Vibes**.

Firmware on the board: handshake (`handshake v160` at the time of writing). BLE name: `ESP32S3 IMU sim`.

Cloud (production): `https://apps.f0xx.org/app/good_vibes`

This is the complete field FAQ: every supported topology, how to prepare hardware, how to tune the phone pipeline, and where each metric lives (app vs Grafana vs raw files).

---

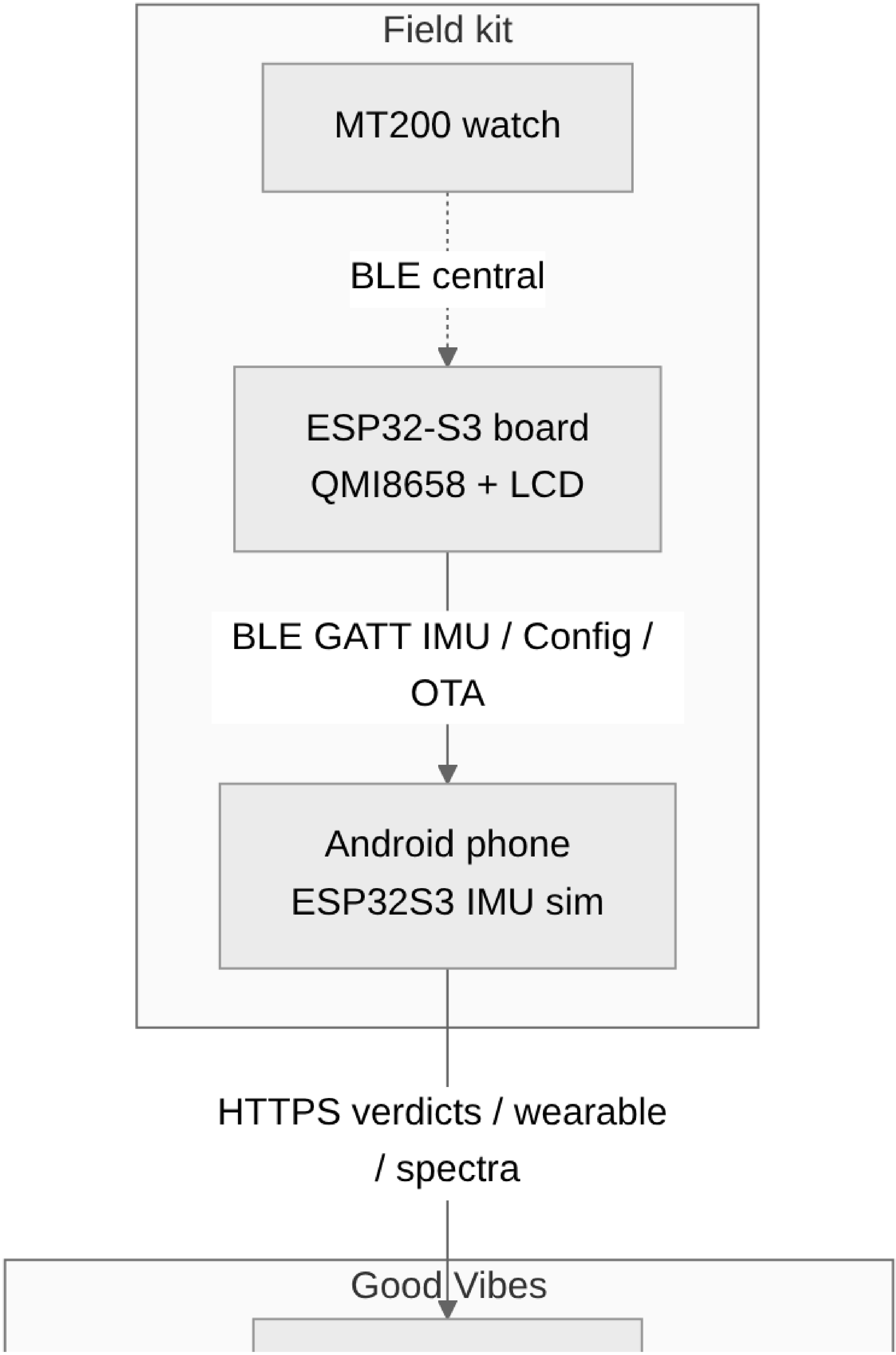
0. Read this first

Rule	Why
The **phone never talks to the MT200**. Only the ESP32 can hold the watch link.	The watch is **single-LE-link**. H-Band XOR ESP32 XOR a laptop.
The **ESP32 never talks to Good Vibes by itself** in the current handshake build.	WiFi STA is compiled **off** (`CONFIG_WIFI=n`). The phone is the relay.
Raw IMU samples stay on the BLE link / phone UI. Cloud stores **verdicts**, **wearable samples**, **spectra (on demand)**, **crashes**, **battery bench**.	Bandwidth and flash.
Glue the **board**, not a flying wire to the IMU. The QMI8658 is on the PCB (I2C SDA 48 / SCL 47).	You measure whatever mechanical path reaches that chip.
After moving the sensor to a new machine, **re-run the reference wizard**.	Old “healthy” fingerprints become false ALERTs.

Topology cheat-sheet

Case	Hardware	Our app?	What you get
A ESP32 + phone	Board + Android	Yes	Live IMU, steps/walk, vibro, AHRS, cloud relay
B ESP32 + MT200 + phone	Board + watch + Android	Yes	Case A **plus** HR / SpO2 / steps from the watch, hop RSSI
C MT200 + phone only	Watch + H-Band	**No**	Use the vendor **H Band** app. ESP32S3ImuSim cannot pair the watch.

Case	Hardware	Our app?	What you get
D ESP32 alone	Board, no phone	Limited	LCD scene, RGB LED, local verdicts; **no cloud**, no watch (bridge needs the debug build, still no HTTP)
E ESP32 + WiFi (no phone)	—	**Not current**	Handshake has WiFi **disabled**. Re-enable only with a coexistence-tested firmware.


Field kit
MT200 watch
BLE central
ESP32-S3 board
QMI8658 + LCD
BLE GATT IMU / Config / OTA
Android phone
ESP32S3 IMU sim
HTTPS verdicts / wearable / spectra
Good Vibes

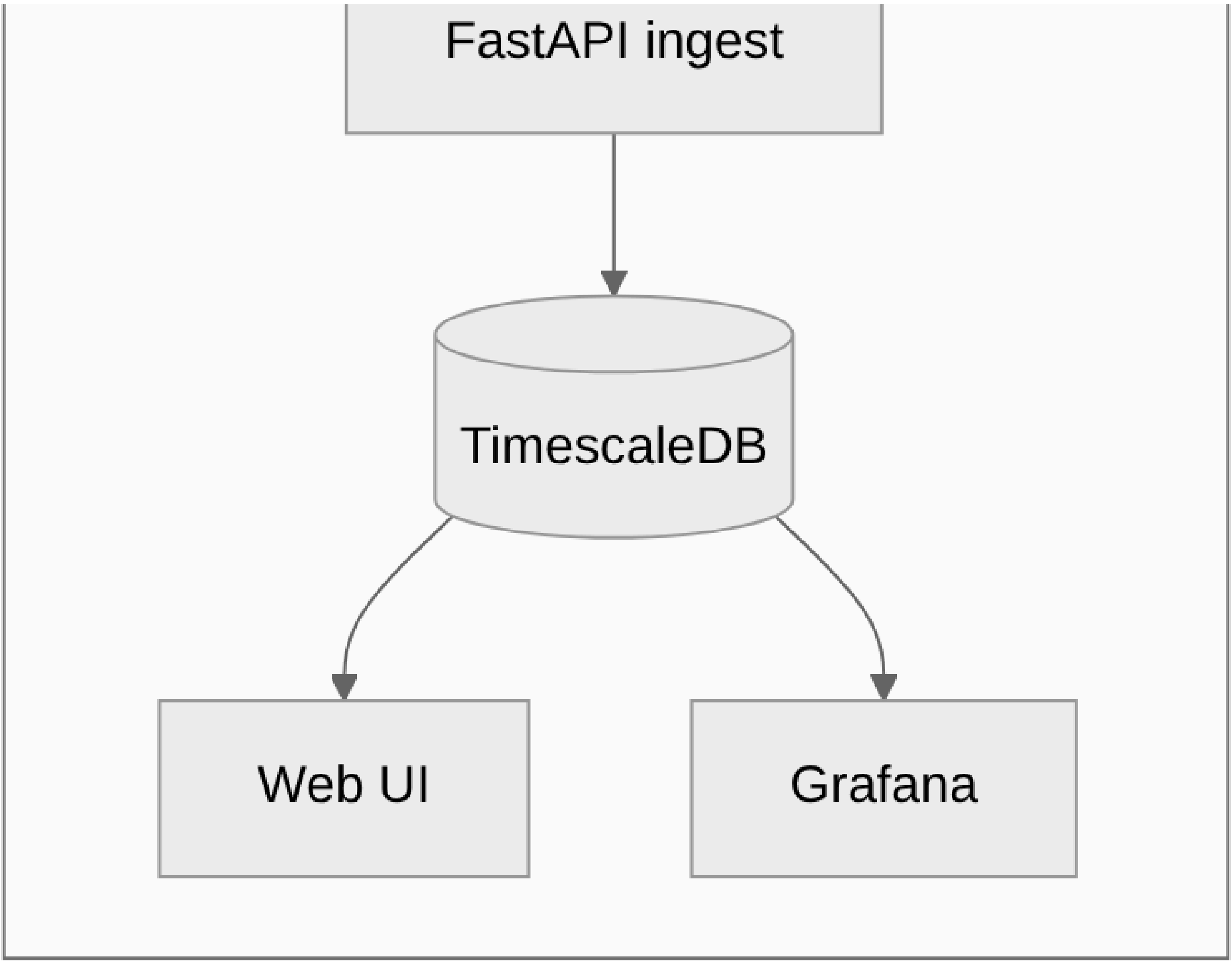


Figure 1

---

1. Shared prerequisites (every use case)

1.1 Hardware checklist

1. Board charged or on USB-C (USB is fine for setup; **unplug USB** for battery-life or vibro-on-machine work).
2. Phone: Bluetooth on, location permission allowed (Android BLE scan), notification permission for the foreground relay.
3. App installed: package `com.esp32s3.imusim`. Test phone in this project is BL6000Pro; any Android 8+ works.
4. Optional watch: MT200 charged, **H Band force-stopped and Bluetooth disconnected from the watch**. If H-Band holds the link, the ESP32 cannot connect.

1.2 First boot — board

1. USB-C to a PC if you need a console; otherwise just power the battery.
2. LCD should show the handshake scene (horizon / live IMU). Serial (115200) should contain `handshake vNN`, `crash ring ready`, `BOOT armed (released=1)`.

3. BLE advertising name: **ESP32S3 IMU sim**.
4. Acrylic edge LED (WS2812 on GPIO38) shows setup / arm / fault / battery — see [Appendix: Acrylic status LED](#).

1.3 First launch — phone

1. Open **ESP32S3 IMU sim**.
2. Grant Bluetooth / nearby devices / notifications when asked.
3. Tap **Connect**. Wait until the status line shows connected (not "Disconnected").
4. Leave the app in the background if you want the **autopilot relay** (periodic BLE sync). Foreground gives the live scene.

1.4 Cloud (do this once if you want dashboards)

1. **Device… → Cloud**.
2. URL: `https://apps.f0xx.org/app/good_vibes` (default).
3. Paste the API key (or **Import from clipboard** / setup link).
4. Device ID: leave default or set a stable name per board (Grafana filters on this).
5. Group ID: e.g. `press-1`, `body-lab`.
6. **Test connection**, then **Upload now** after you have data.
7. Bridge mode: leave **Rendezvous** unless you are lab-debugging (then shorter interval).

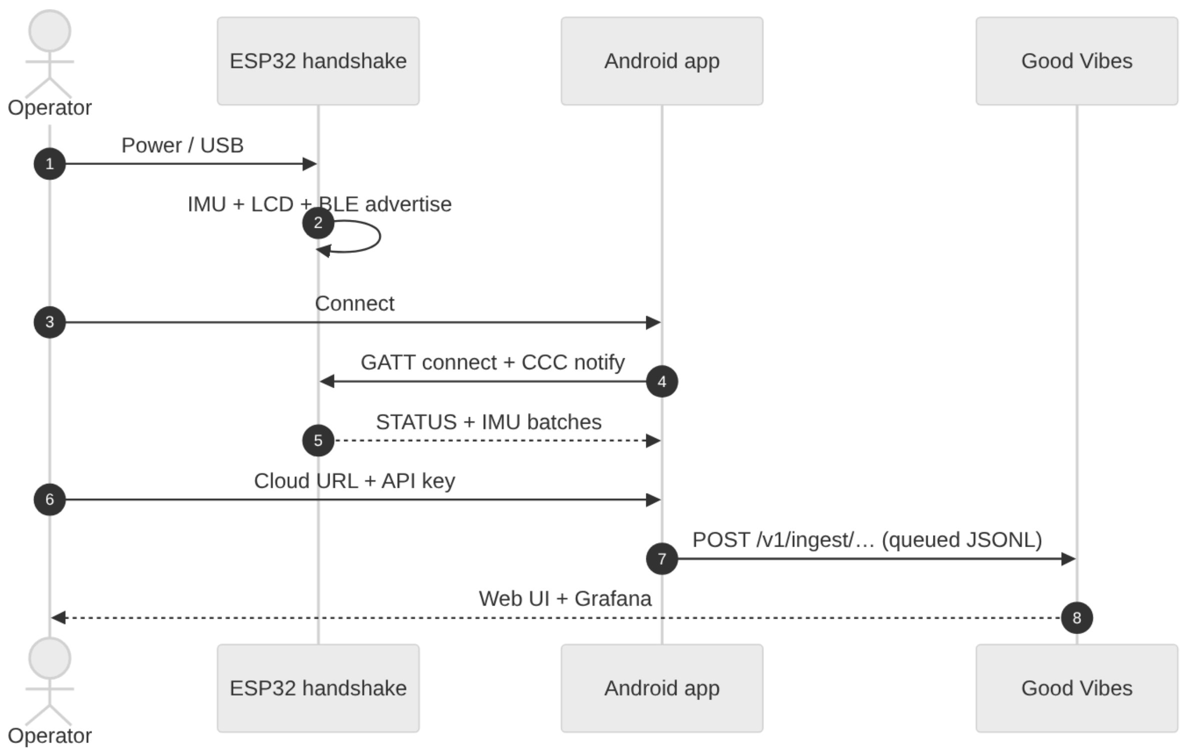


Figure 2

1.5 Where to look — three layers

Layer	What it is	When to use
Phone UI	Live scene, vibro caption, banners	Tuning, "is it alive?"

Layer	What it is	When to use
Good Vibes web	`…/app/good_vibes/` verdicts; `…/wearable` live HR/steps	Shift overview, CSV/TSV export
Grafana	Verdicts, wearable, crashes, battery bench	Trends, lag, RSSI, edge features

Production Grafana (behind the same host): `https://apps.f0xx.org/app/good_vibes/grafana/`

Dashboard	UID	Use
ESP32 IMU Verdicts	`imu-verdicts`	RMS, corr, bands, edge, battery, ingest lag
Wearable (MT200)	`imu-wearable`	HR, SpO2, steps, walk_cm, hop RSSI, silent gaps
ESP32 IMU Crashes	`imu-crashes`	Fault ring uploads
Battery bench	`imu-battery-bench`	Discharge sessions

---

2. Body health measurements

Two hardware variants. Software path is the same on the phone; the watch only adds PPG / step counter samples.

2.1 Variant A — ESP32 only (no watch)

What the board measures: 6-axis IMU (accel + gyro). Firmware derives **walk distance / steps-like walk_cm**, attitude, and live scene. There is **no optical HR or SpO2** on the ESP32.

What you watch for: motion, gait/walk_cm, orientation, battery, chip temperature.

2.2 Variant B — ESP32 + MT200 (recommended for HR / SpO2)

What the watch measures: heart rate, SpO2, on-watch step counter, watch battery.

What the ESP32 adds: IMU walk_cm, BLE RSSI of the watch hop, relay to the phone.

Hard rules

- Wear the watch **snug, optical window on skin**. Table / no contact → HR stays `0` for ~30 s then still 0 (not a firmware bug).
- HR lock takes **~30 s** on-wrist (`D0` frames). SpO2 is **mutually exclusive with HR** (shared PPG). Firmware cycles them; do not expect both at 1 Hz.
- **Quit H Band** completely before powering the ESP32 bridge.
- Wrist RSSI is often −70 dBm on a table and −80…−86 dBm on-wrist. First connect can fail with RF noise; the bridge retries.

Current handshake merges crash-debug extras by default (`CRASH_DEBUG=1`), which **includes the MT200 central bridge** and autostart after BLE is up.

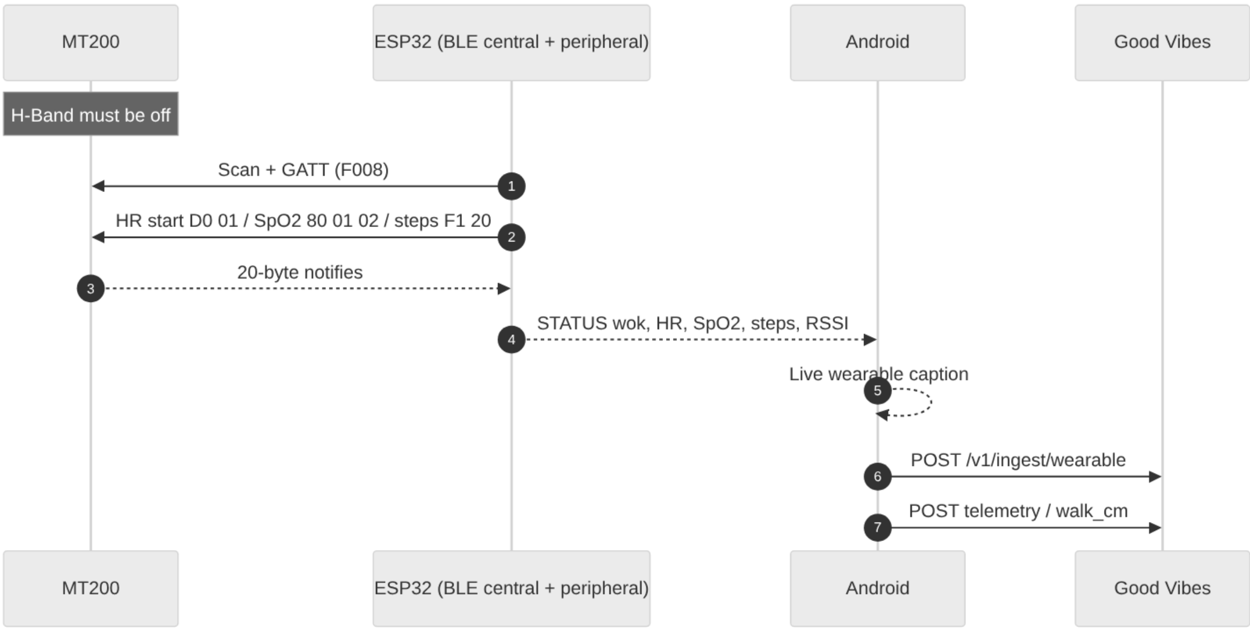


Figure 3

2.3 Preparations (body)

1. Charge board + watch. Confirm LCD alive.
2. Strap the board so the **IMU chip side** is toward the body segment you care about (waist / ankle / chest pack). Avoid a floppy lanyard — the QMI8658 will report strap bounce as motion.
3. Optional: **Device… → Floor level calibration** on a known-level table if you care about tilt, not if the board is worn loosely.
4. Phone: Connect BLE.
5. **Device… → Profile wizard** → **Body sensor (computed)** (live IMU + steps, performance power) **or Body sensor (scene mirror)** if you want the LCD/scene mode mirrored.
6. Cloud as in §1.4. Group ID e.g. `body`.
7. For watch: put MT200 on wrist, kill H-Band, wait ~1 min after ESP boot for HR lock.

2.4 Phone pipeline (body)

Step	Menu / control	Setting
1	Main **Connect**	Stay connected or rely on autopilot
2	Mode chips	**Computed IMU** for gait/steps; **Angles / scene** to watch the LCD horizon
3	Profile wizard	Body computed / body scene
4	Cloud	Enabled, device_id unique per wearer or per board
5	Do **not** open Vibro reference wizard	That path is for machines
6	Optional **AHRS live view**	Full CPU 240 MHz + 100 Hz IMU — **not** a battery saver; yaw drifts (no magnetometer)

2.5 Metrics — attention, dashboard, raw

Metric	Meaning	Healthy-ish	Phone	Dashboard	Raw
HR	Optical bpm	40–180 locked; 0 = no lock	Wearable live page after upload	Grafana **Wearable**	`wearable_samples` kind= `hr`
SpO2	%	95–100 on-wrist; `1` on wire = wear fail, not 1%	Wearable page	Grafana Wearable	kind= `spo2`
Watch steps	Watch pedometer	Monotonic during walk	Wearable page	Grafana Wearable	kind= `steps`
IMU `walk_cm`	Board integrated walk	Rises when the **board** moves	STATUS / wearable page	Grafana Wearable (compare vs watch steps)	kind= `walk_cm`
ESP↔phone RSSI	Phone-measured	Roughly −40… −80 indoor	STATUS	Wearable RSSI panels	ingest rssi fields
MT200↔ESP RSSI	ESP-measured	−70 table, worse on wrist; −127 = N/A	STATUS	Wearable hop RSSI	`mt200` rssi
Battery % / V	Board LiPo	Trend down off USB	Main caption	Verdicts + wearable	STATUS `pct` , `v`
Chip temp	SoC	Rises in AHRS / sun	STATUS	Verdicts	`tc`

Do not treat IMU walk_cm and watch steps as the same counter. They disagree on purpose (different sensors). Use them as a **cross-check**: watch steps up + walk_cm flat ⇒ board isn't on the walking limb.

Live web: `https://apps.f0xx.org/app/good_vibes/wearable`

CSV: Good Vibes **Download CSV** / Copy TSV (delivery lag included on new ingest).

---

3. Vibration analysis (machines)

This is the industrial path: edge FFT / band RMS **on the ESP32**, phone as relay, cloud verdicts.

3.1 What you are actually measuring

The QMI8658 sits on the **PCB**, near the I2C header (SDA GPIO48, SCL GPIO47). You are **not** measuring a bearing race directly unless the mechanical path is stiff.

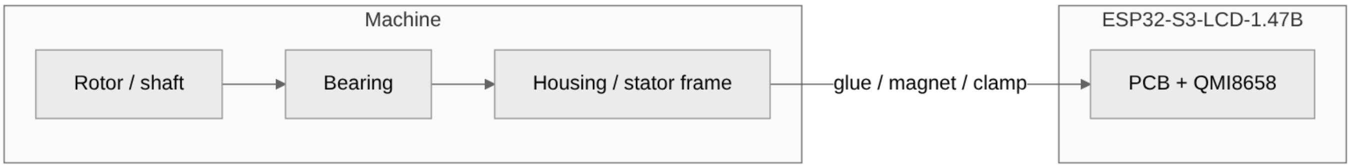


Figure 4

Target	Where to glue / clamp	Axis hint
Rolling bearing (radial)	Bearing housing, load zone if known; else drive-end cap	Radial = across shaft; keep board flat on housing
Thrust / axial	Housing face along shaft centreline	IMU X/Y in the plane of the board; note which way the USB points and **write it on the machine**
Rotor unbalance	Rigid housing, **not** a guard sheet-metal	Same as radial bearing
Stator / frame	Motor foot or stator yoke, away from cooling fans that flap	Often lower frequency; use **Low RPM** or **Ultra-low RPM** preset
Gearbox	Split line or bearing cap, not the inspection cover	
Pump / fan	Pump volute or fan pillow block	Prefer **Intermittent** if the machine is not 24/7

Mounting rules

1. **Stiff path:** cyanoacrylate on a cleaned paint-free pad, or a strong magnet on a steel housing. Foam tape is for demos only (it low-pass-filters).
2. **One orientation forever** after the reference wizard. Rotating the board 90° invalidates fingerprints.
3. **Do not** mount on a vibrating cable tray or the plastic LCD bezel as the only contact — couple the **back of the PCB / metal can** to the machine.
4. LCD can face out for a glance; the IMU still needs the stiff path.
5. Keep USB cable off the machine while capturing (cable slap looks like an ALERT).

3.2 Preparations (vibro)

1. Machine in **normal idle / normal production** — not a crash-stop, not a cold start unless that *is* the reference.
2. Glue/clamp as above. Mark USB direction with a paint pen.
3. Phone BLE connect. Unplug USB from the board if you care about battery schedule / deep sleep.
4. **Device… → Profile wizard** — pick a vibro preset (table below).
5. **Vibro… → Reference wizard** on **this** mounting. Confirm erase if the sensor moved.
6. Cloud group_id = machine name. Device ID = this board.

3.3 Phone pipeline (vibro)

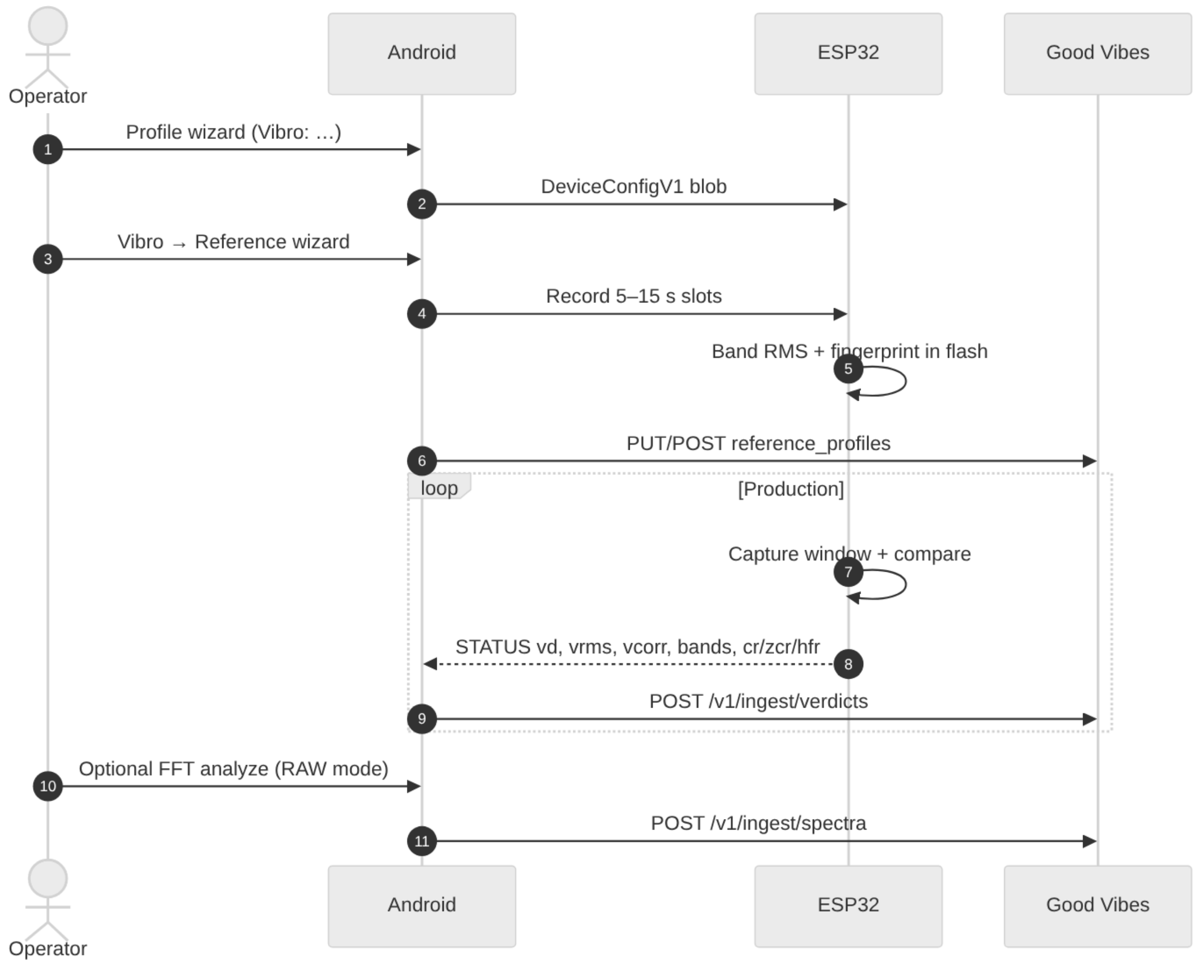


Figure 5

Preset (Profile wizard)	When	Capture idea
Vibro: normal	Typical 50/60 Hz machines, continuous	~15 s window / 60 s interval, 100 Hz IMU
Vibro: low RPM diagnosis	Slow shafts, more time per rev	30 s / 120 s, 50 Hz
Vibro: ultra-low RPM	Very slow / high inertia	60 s / 5 min, 25 Hz
Vibro: intermittent machine	Starts/stops; save battery	Hourly random slot, deep sleep, mix windows
Vibro: machine monitor (mix)	Unattended monitor	Intermittent + mix/dyn sub-windows

Then:

1. **Vibro… → Reference wizard…**

- Erase slots if this is a new machine.

- Choose 1–5 references (3 is a good default).

- Each take: **normal idle vibration ~12 s**, hold still relative to the housing (you are not “shaking” a turbine).

- **Upload & finish**.

1. **Vibro… → Vibro mode** to switch diagnosis tier later without the full wizard.
2. **Vibro… → Capture mix…** (expert) only if you know mix/dyn ratios.
3. **Vibro… → FFT analyze** switches the ESP to **Raw IMU**, collects a burst, uploads a 128-bin spectrum. Use when Grafana verdicts look wrong and you want a picture of the line.
4. **Vibro… → Verdict history** — last on-phone levels without opening Grafana.

Acrylic LED (GPIO38): **red** solid on WARN/ALERT (and other NOK); **off** when armed and healthy. Colour chart: [Appendix: Acrylic status LED].

3.4 Metrics — attention, dashboard, raw

Metric	Meaning	Attention	Phone	Dashboard	Raw
`vd` level	0 OK, 1 WARN, 2 ALERT	ALERT after a good reference ⇒ process change or loose mount	Vibro caption	Grafana **Verdicts**	`verdicts.level`
`vrms` / `vpeak`	Time-domain energy	Step change after maintenance	Caption	Verdicts RMS panel	columns
`vcorr` / `bcorr`	Similarity to reference	Drop ⇒ spectral shape changed	Caption	band_corr panel	`raw_json`
`bdmax`	Worst band delta	Which band moved	Caption	band_delta_max	`raw_json`
`bnd[]` / `b16[]`	4 band RMS	Compare to ref bands	—	band time series	`raw_json.bands`
`cr` crest	Peakiness	Impacts, looseness	Caption	edge_crest	`raw_json`
`zcr`	Zero-crossing rate	High-frequency hash	Caption	edge_zcr_hz	`raw_json`
`hfr`	High-frequency ratio	Bearing-ish vs 1× running speed	Caption	edge_hf_ratio	`raw_json`
`edge_score` / `edge_risk`	**Server** heuristic	Trend, not a trip	—	Verdicts	computed on ingest
Spectrum bins	Phone FFT of RAW	“What frequency?”	FFT toast	(API spectra)	`POST /v1/ingest/spectra`
LED	Local	Red = act now	Eyes	—	—

Live web: `https://apps.f0xx.org/app/good_vibes/`

Do not stare at live RAW IMU in Grafana — it is not stored. Use FFT upload or the phone scene.

False ALERT checklist: loose glue, USB cable, reference taken while the machine was off, board rotated, nearby hammering, deep-sleep preset on a machine that should be continuous.

---

4. Other operator workflows

4.1 Floor / mounting calibration

Device… → Floor level calibration…

- Purpose: persistent tilt correction vs a **true level** (bubble). Not a vibration reference.
- Place board on a known-flat surface, still, **Start**.
- Clear if you remount at a new angle.

4.2 AHRS / orientation lab

Device… → AHRS live view (full speed)…

- Complementary filter, gyro+accel, **no magnetometer** → yaw drifts.
- Forces 240 MHz + 100 Hz. Leave the screen to restore Auto.
- 3D debug page on the backend: `ahrs.html` (lab).

4.3 Battery bench

Device… → Battery bench…

- Unplug USB. Start. Config locks until Stop.
- Samples upload when cloud is on. Grafana **Battery bench**.

4.4 WiFi wizard (provisioning only)

Device… → WiFi wizard still talks **over BLE**. Handshake **does not bring WiFi STA up** in the current `prj.conf`. Use it only if you are on a firmware that re-enabled WiFi. Hotspot fallback (when implemented on that build): `ESP32-IMU-Setup` / `imu12345` → `http://192.168.4.1`.

4.5 OTA (app then firmware)

1. Phone and board connected (or autopilot will pick up later).
2. **Device… → Check for OTA**.
3. **APK first**, then firmware. The phone refuses FW OTA if the app is too old (`min_apk_versionCode`).
4. USB `flash-zephyr.sh handshake` remains the recovery path if BLE OTA fails.

4.6 Crash debug (lab)

Device… → Crash debug (dev)… only on builds with `dbg=1`. Injects a fault; relay uploads the crash ring. Grafana **Crashes**. Do not use on a production machine capture.

5. Topology deep-dive (FAQ of combinations)

5.1 ESP32 + phone (no watch)

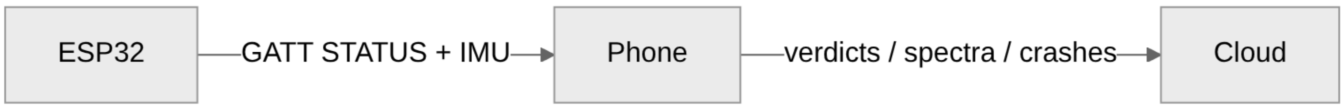


Figure 6

Use: body IMU, vibration, AHRS, battery bench.

You will not see: HR, SpO2, watch steps. Wearable page may still show `walk_cm` from the IMU.

5.2 ESP32 + MT200 + phone

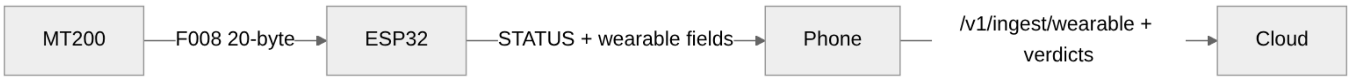


Figure 7

Use: body health with optical sensors **and** optional vibro if the board is also on a machine (unusual — pick one mounting).

Setup extra: H-Band off; wait for HR lock; compare watch steps vs `walk_cm`.

5.3 MT200 + phone (no ESP32)

Not supported by ESP32S3 IMU sim. Use **H Band**. Our cloud never sees those samples unless some other bridge exists.

5.4 ESP32 + MT200, no phone

Watch samples sit in ESP RAM/STATUS only. **No Good Vibes**. LCD will not show a medical dashboard. Bring a phone for ingest.

5.5 Two phones

Only one BLE central should hold the ESP32. A second phone can open Good Vibes in a browser (cloud), not a second BLE connection.

5.6 Autopilot / rendezvous (unattended vibro)

Phone in a locker near the machine:

1. Cloud on, bridge **Rendezvous**.
2. Intermittent / machine-monitor preset so the ESP sleeps between windows.
3. Phone must stay powered and in BLE range. WorkManager uploads when the WAN is up.

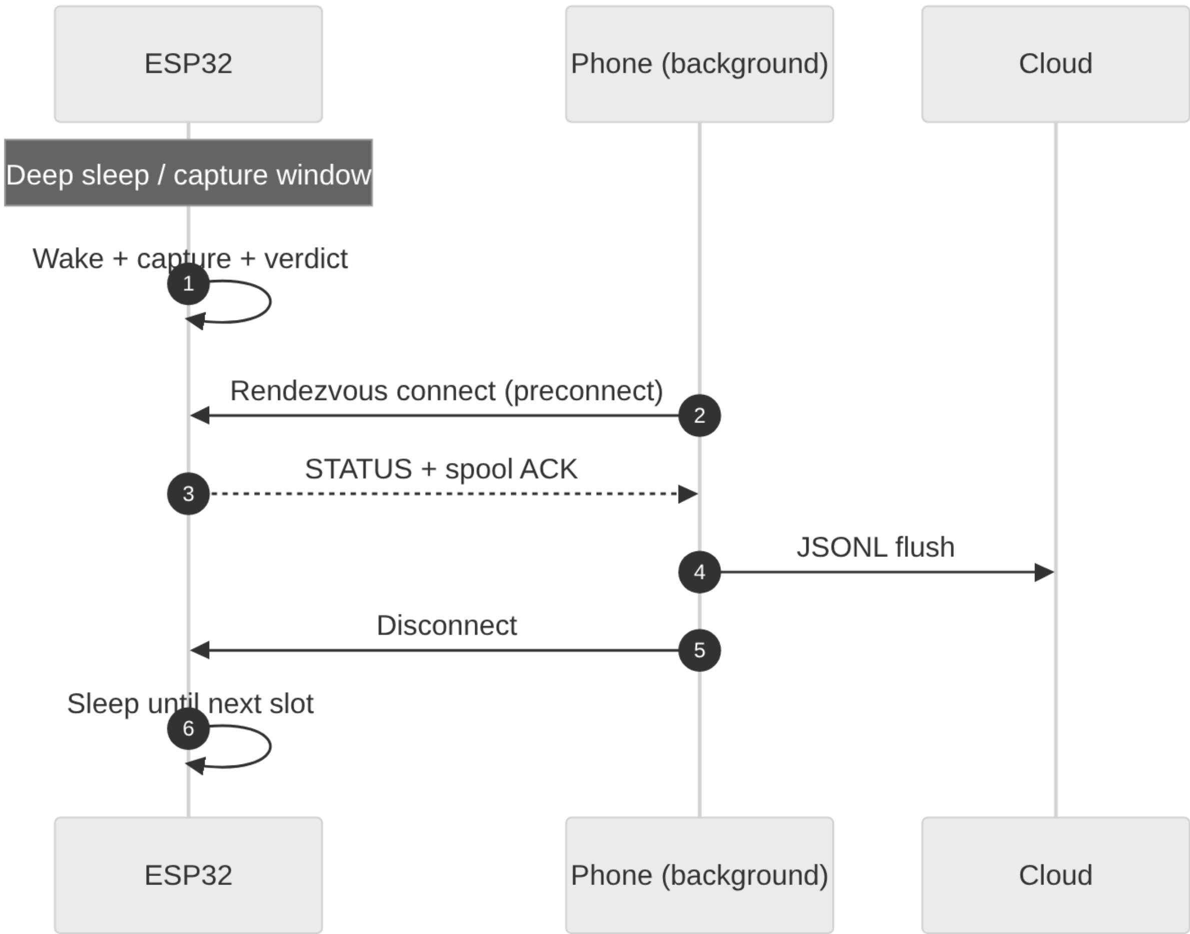

ESP32
Phone (background)
Cloud
Deep sleep / capture window
Wake + capture + verdict
1
Rendezvous connect (preconnect)
2
STATUS + spool ACK
3
JSONL flush
4
Disconnect
5
Sleep until next slot
6
ESP32
Phone (background)
Cloud

Figure 8

6. Data analysis — from prep to output

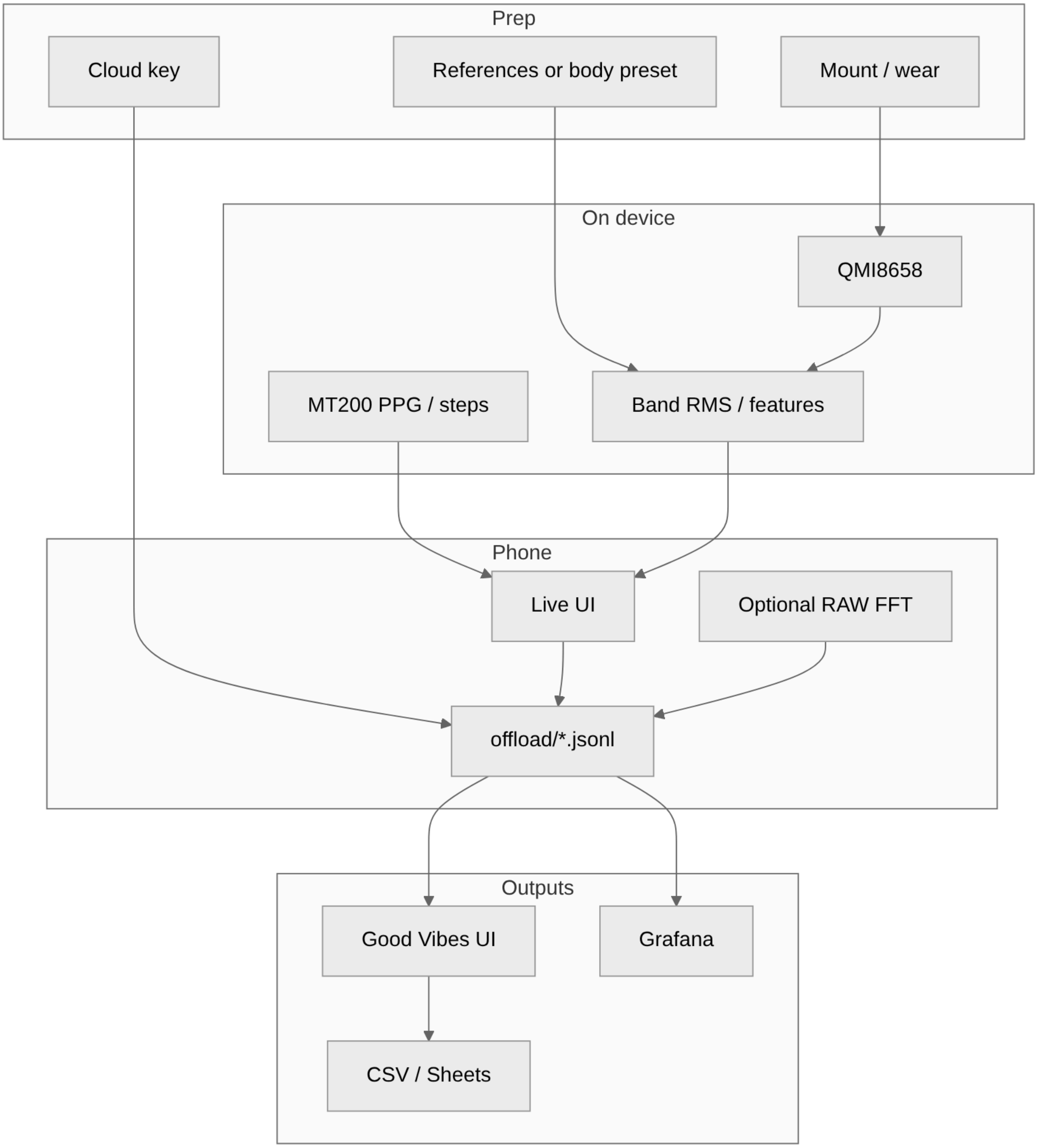


Figure 9

Recommended daily loop

1. Grafana Wearable or Verdicts — last 24 h, look for **holes** (relay down) vs **ALERT clusters** (real process).
2. Good Vibes table — CSV if you need Sheets.
3. Phone only when Grafana is ambiguous (FFT, ref wizard, LED).

Export: Good Vibes **Download CSV** joins wearable + telemetry + verdicts + ingest batches (`lag_ms` , RSSI, bytes). Historical rows may have empty `delivered_at` until new ingest.

---

7. Troubleshooting FAQ

Connect button does nothing / scan empty

Bluetooth + location permission; board advertising; another phone already connected.

Connected but no IMU

Wait 2 s for CCC; reboot board; confirm handshake not `smoke` firmware (`WS147B-Zephyr` is smoke-only).

Watch never locks HR

H-Band still connected; watch not on skin; wait 30+ s; RSSI −90 or −127; crash-debug/MT200 not in this image (`CRASH_DEBUG=0`).

SpO2 is 1%

Wear-fail flag, not saturation. Tighten strap. Stop expecting HR in the same second.

ALERT with a quiet machine

Bad or empty references; mount loose; USB cable; board moved. Re-wizard.

Grafana empty, phone live

Cloud key/URL; **Upload now**; WAN; device_id mismatch vs Grafana variable.

Two data holes: wearable vs verdicts

Different pipelines. Wearable is fire-and-forget; verdicts retry via JSONL. A quiet STATUS does not mean the watch ingest died.

OTA offered forever

Phone APK still older than manifest `min_apk_versionCode` , or FW_VERSION_CODE not bumped on the image you flashed.

Cast builder vs IMU builder (lab)

Android Cast and IMU are **separate tenants**. IMU jobs must show project `imu` , containers `imu-bld-*` / `imu-zephyr-bld-*` , channel `imu` . Never `androidcast-bld-*` , never Cast OTA version `00.01.00.xxxx` . See `ci/cast/README.md` .

What does the acrylic LED colour mean? (board status / readiness)

Single WS2812 on GPIO38. ✓ = that RGB channel is on. Empty = off. `*reserved*` = that colour combo is not wired. Flash is 2 s on / 2 s off. Healthy armed run is **off**. Full date/FW wiring: [Appendix: Acrylic status LED](#).

Condition	R	G	B
Setup — no reference profiles (solid blue)			✓
Await arm — refs recorded, not started (blue flash 2s/2s)			✓
OK pulse — NVS/config saved (~2 s solid blue)			✓
NOK — WARN/ALERT or armed without loaded ref (solid red)	✓		
Battery ≤10% SOC on battery, not USB/DC (yellow flash 2s/2s)	✓	✓	
Operational — armed, refs OK, no fault (off)			
Green only	*reserved*	*reserved*	*reserved*
Magenta (R+B)	*reserved*	*reserved*	*reserved*
Cyan (G+B)	*reserved*	*reserved*	*reserved*
White (R+G+B)	*reserved*	*reserved*	*reserved*

---

8. Quick recipes

Body + watch, 10 minutes

Kill H-Band → power ESP → Connect phone → Profile **Body sensor (computed)** → Cloud on → wear watch → wait for HR → open `/wearable`.

New pump, 20 minutes

Glue on bearing cap → Connect → Profile **Vibro: normal** → Reference wizard ×3 at idle → Cloud group `pump-7` → confirm LED **off** after arm (healthy run) → Grafana Verdicts.

Overnight unattended

Intermittent or machine-monitor preset → Rendezvous bridge → phone plugged in BLE range → morning Grafana + CSV.

---

9. File / URL index

Thing	Where
App package	`com.esp32s3.imusim`
BLE name	`ESP32S3 IMU sim`
Cloud	`https://apps.f0xx.org/app/good_vibes`
Wearable UI	`…/wearable`

Thing	Where
Grafana	`…/grafana/`
Phone queue	app files `offload/verdicts.jsonl` (+ gzip)
Firmware flash	`zephyr/scripts/flash-zephyr.sh handshake`
Acrylic LED colours	Appendix: Acrylic status LED
MT200 protocol notes	veepoo-proto-ble-reverse.md
Metrics schema	metrics.md
Architecture	architecture.md

Appendix: Acrylic status LED

Wired: 2026-08-23 (handshake `vibro_led.c` schema in commit `7de0208`).

Firmware: `handshake v160` (`FW_VERSION_NAME` / `FW_VERSION_CODE` 160 in `zephyr/app/common/fw_version.h`).

The red acrylic edge light is a **single WS2812** pixel on **GPIO38** (GRB wire order). Channel drive is 48/255 when on. Handshake turns the pixel **off** at boot, then `vibro_led` owns it for the rest of the run. The diffuser is red: a firmware **green** channel looks red through the plastic, so the "OK pulse" is driven as **blue**, not green.

Solid vs flash. Solid means the channel stays on. Flash is 2 s on / 2 s off (`LED_FLASH_PERIOD_MS` 4000, `LED_FLASH_ON_MS` 2000). The OK pulse is a **solid ~2 s** burst (`LED_OK_PULSE_MS`), then the base state returns. Flash off-phases look the same as operational off.

Priority (first match wins): OK pulse → battery critical → setup (no refs) → await arm → NOK → off.

✓ = that RGB channel is driven for the condition. Empty = channel off. `*reserved*` = that colour combo is **not wired** in this firmware.

Condition	R	G	B
Setup — no reference profiles in flash (solid blue)			✓
Await arm — refs recorded, not started / CMD 10 (blue flash 2s/2s)			✓
OK pulse — NVS / config saved (~2 s solid blue)			✓
NOK — verdict WARN/ALERT, or armed without a loaded reference (solid red)	✓		
Battery critical — ≤10% SOC **and** on battery, not USB/DC (yellow flash 2s/2s)	✓	✓	
Operational — armed, refs OK, no active fault (off)			
Green only	*reserved*	*reserved*	*reserved*
Magenta (R+B)	*reserved*	*reserved*	*reserved*

Condition	R	G	B
Cyan (G+B)	*reserved*	*reserved*	*reserved*
White (R+G+B)	*reserved*	*reserved*	*reserved*

Not a colour in this table. BLE advertising, BLE connected, charging / on USB, OTA, crash, MT200 watch link, and WiFi do **not** set the acrylic LED. Charging specifically suppresses the yellow battery flash (`on_dc` → not critical). Smoke firmware (`WS147B-Zephyr`) only turns the pixel off; it does not use this schema.